



Real-Time Crypto Market Anomaly Detection & Dashboard

Spotting cheating in live crypto trading — shown on a dashboard in seconds

GitLab Repository: <https://gitlab.inf.unibz.it/kodanda.challa/rtbdp-crypto-anomaly>

GitHub Repository: https://github.com/kodandachalla/RTBDP_Project

THE PROBLEM

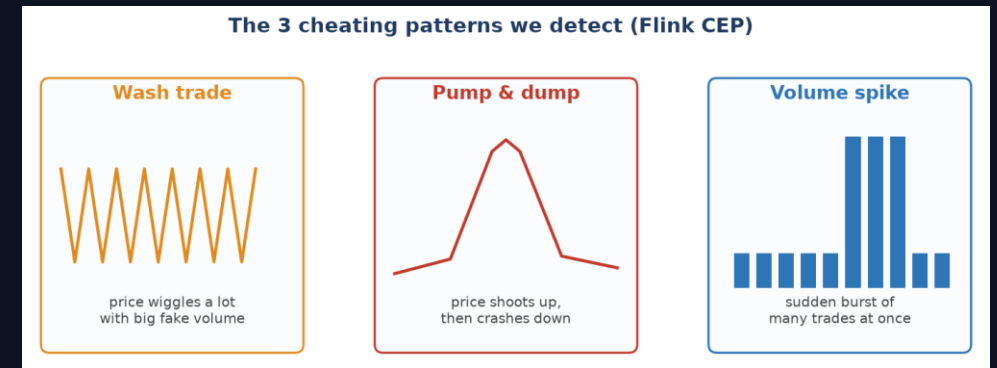
Crypto markets are a target for cheating

Crypto trades 24/7 and produces thousands of trades a second.

Because it is lightly regulated, people cheat:

- Wash trading — a person quietly buys and sells to themselves to make a coin look busier than it really is.
- Pump and dump — a group pushes a price up fast, then sells at the top, leaving buyers with losses.
- Volume spikes — a sudden flood of trades, which can come before a price crash.

Finding these tricks a day later is useless. To help an exchange or regulator the warning must appear within seconds. That is exactly what this project does.



Wash trade: price wiggles + fake volume.

Pump & dump: up fast, then crash.

Volume spike: sudden flood of trades.

Each match is saved as a MEDIUM or HIGH alert and appears on the dashboard in seconds.

THE DATA

Live trades from Binance — free, no key

One always-open connection gives us every trade for all six coins.

`symbol · price · volume · exact time (to the millisecond)`



Fast never stops — 50 to 200 trades per second, per coin



Enough hundreds of trades a second in total — real stream load



A bit messy trades can arrive slightly out of order — we handle that



Tiny each message is small and simple to read

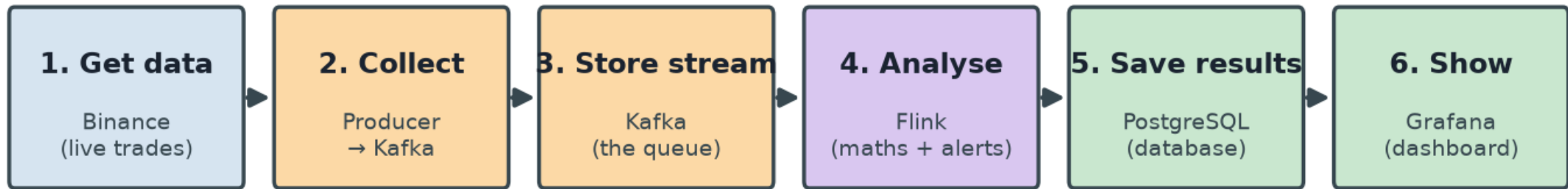
6

coins watched live

Symbol	Name
BTCUSDT	Bitcoin
ETHUSDT	Ethereum
BNBUSDT	BNB (Binance Coin)
SOLUSDT	Solana
DOGEUSDT	Dogecoin
XRPUSDT	XRP (Ripple)

What it does, in six simple steps

What the system does, in 6 simple steps

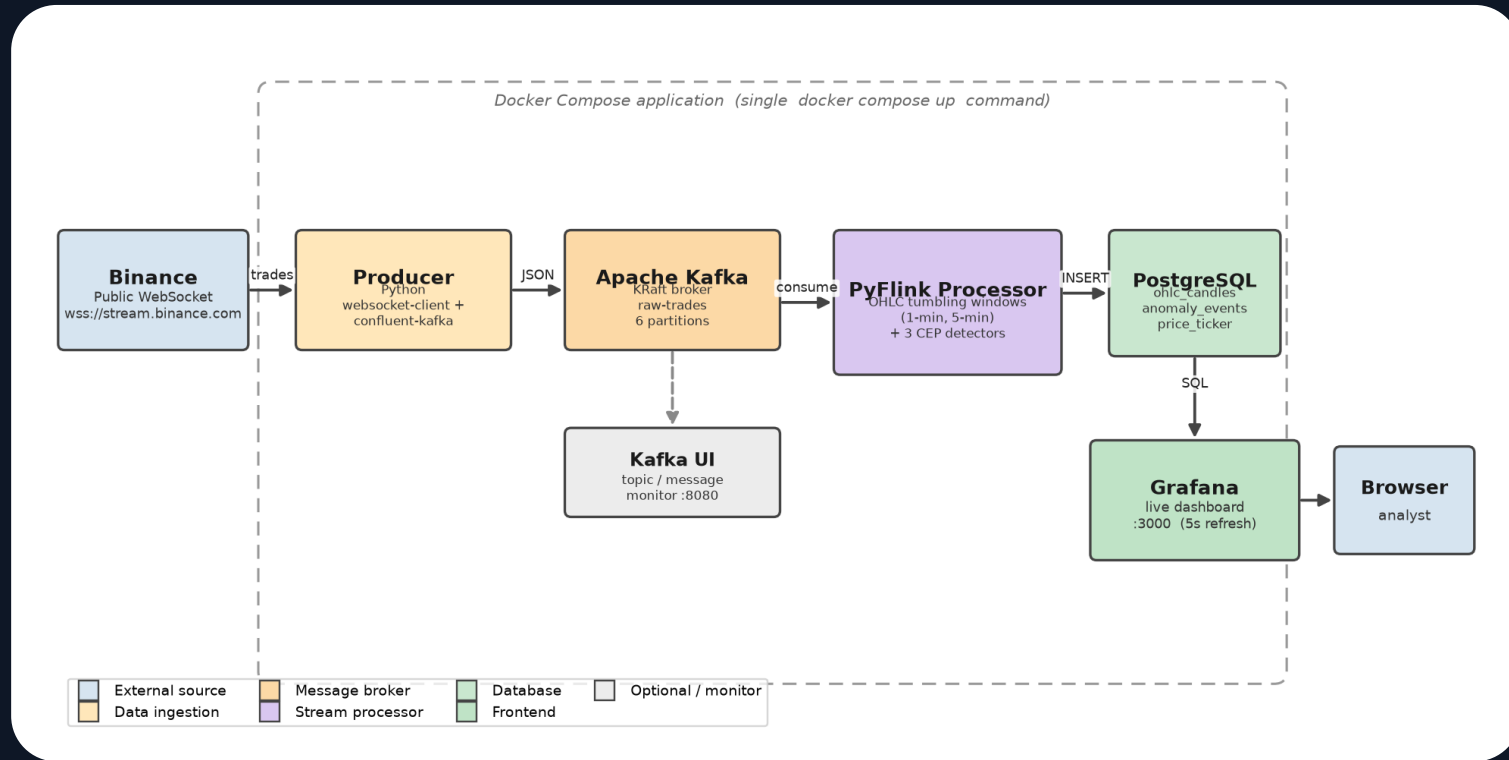


Live trades come in on the left and turn into a live dashboard on the right.

Live trades come in on the left and become a live dashboard on the right. Each step is a small, well-known tool.

HOW IT FITS TOGETHER

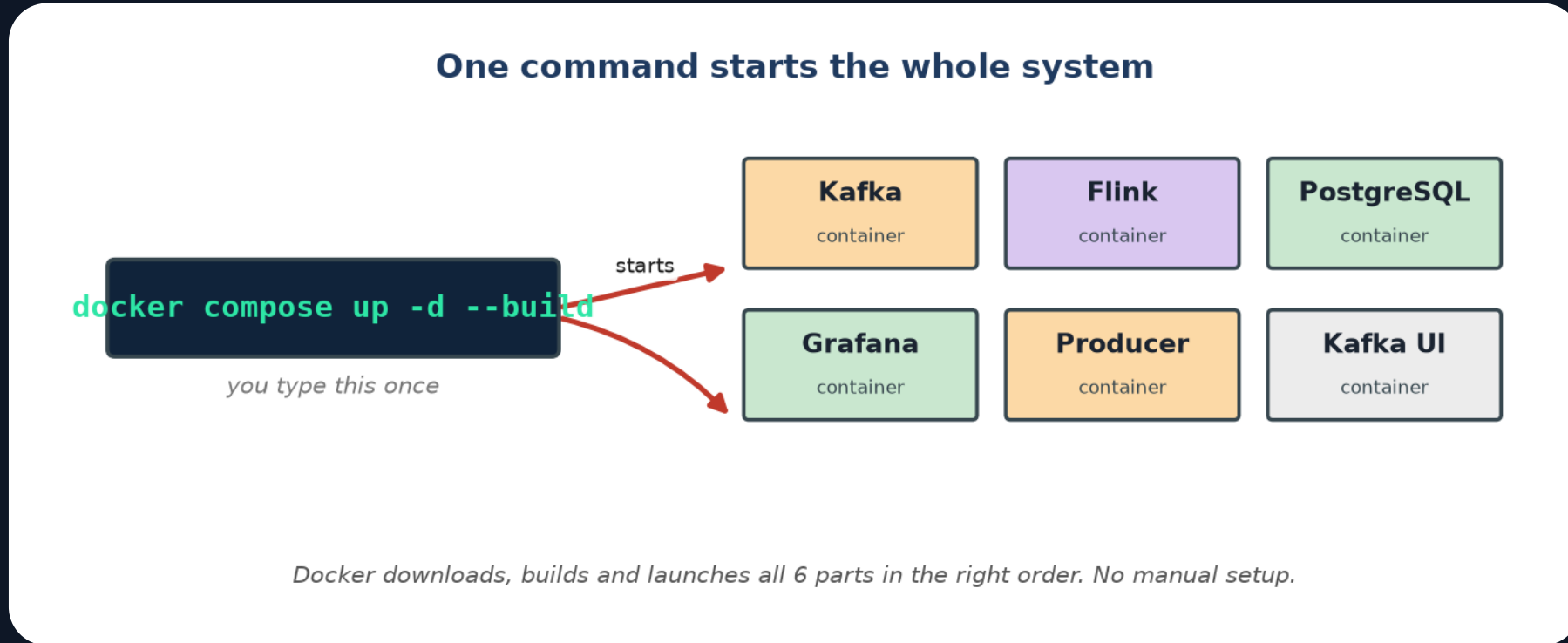
The architecture



Standard course shape: collect → Kafka → Flink → database → dashboard. Every box is one Docker container; they find each other by name.

THE BEST PART

One command starts everything

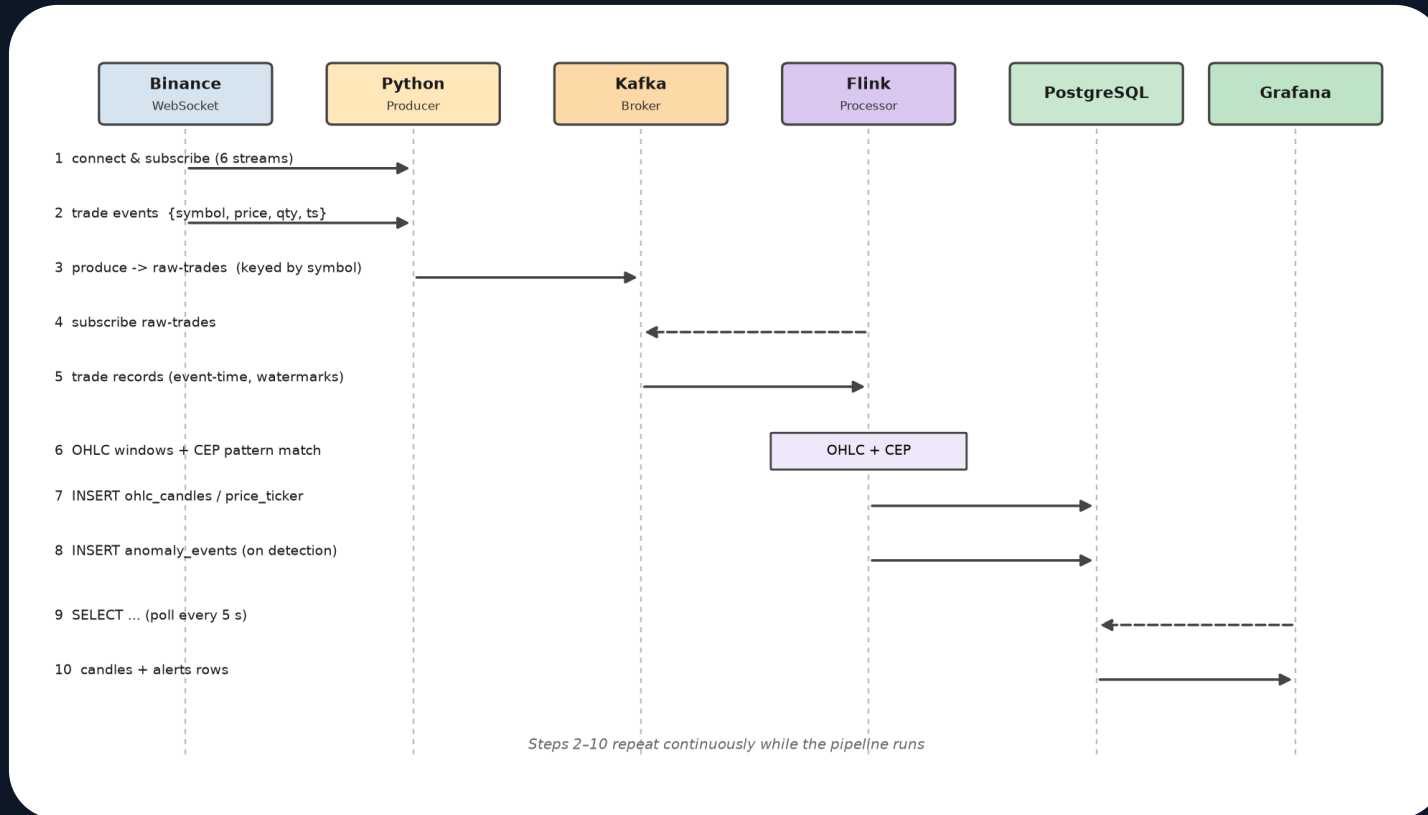


You only install Docker. Then `docker compose up -d --build` downloads, builds and starts all six parts in the right order — no manual setup.

kafka + postgres → kafka-ui + grafana + producer → flink-processor

STEP BY STEP

How one trade travels through the system



1-2

Get the trade

Binance sends a trade. The Python collector reads it and cleans it up.

3

Put in Kafka

The collector puts the trade into Kafka on the lane for that coin.

4-5

Flink reads it

Flink takes the trade using event timestamp and assigns a watermark.

6

Candles + checks

Flink updates the price candles and feeds the trade to all three CEP detectors — each updating that coin's state.

7-8

Save results

Flink saves the candle and any alert into PostgreSQL. The price ticker is upserted.

9-10

Dashboard updates

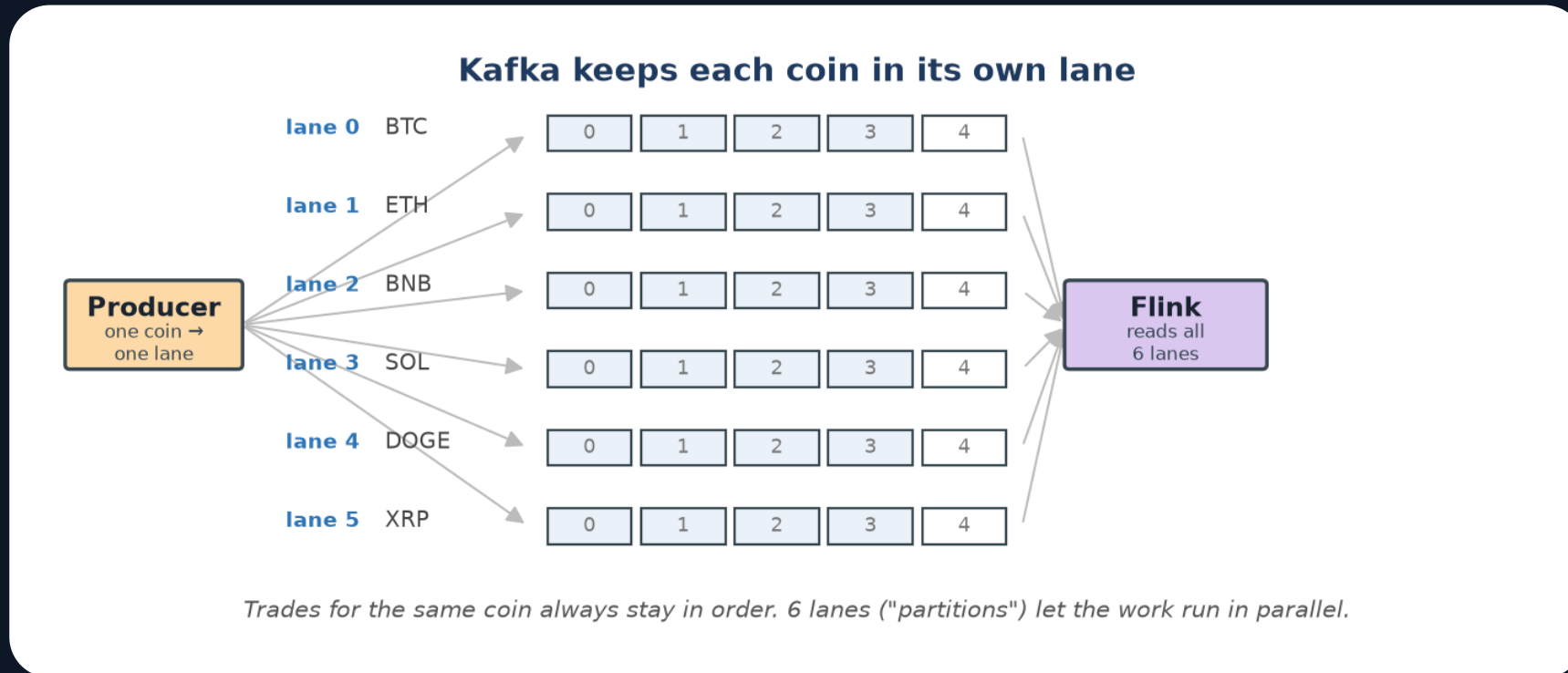
Grafana asks the database for fresh data every 5 seconds and redraws all panels.

Get trade → put in Kafka → Flink reads it → make candles + check patterns → save to database → dashboard shows it. This repeats forever.

Every 30 seconds Flink checkpoints state to disk so a crash resumes cleanly.

Kafka

Each coin gets its own lane

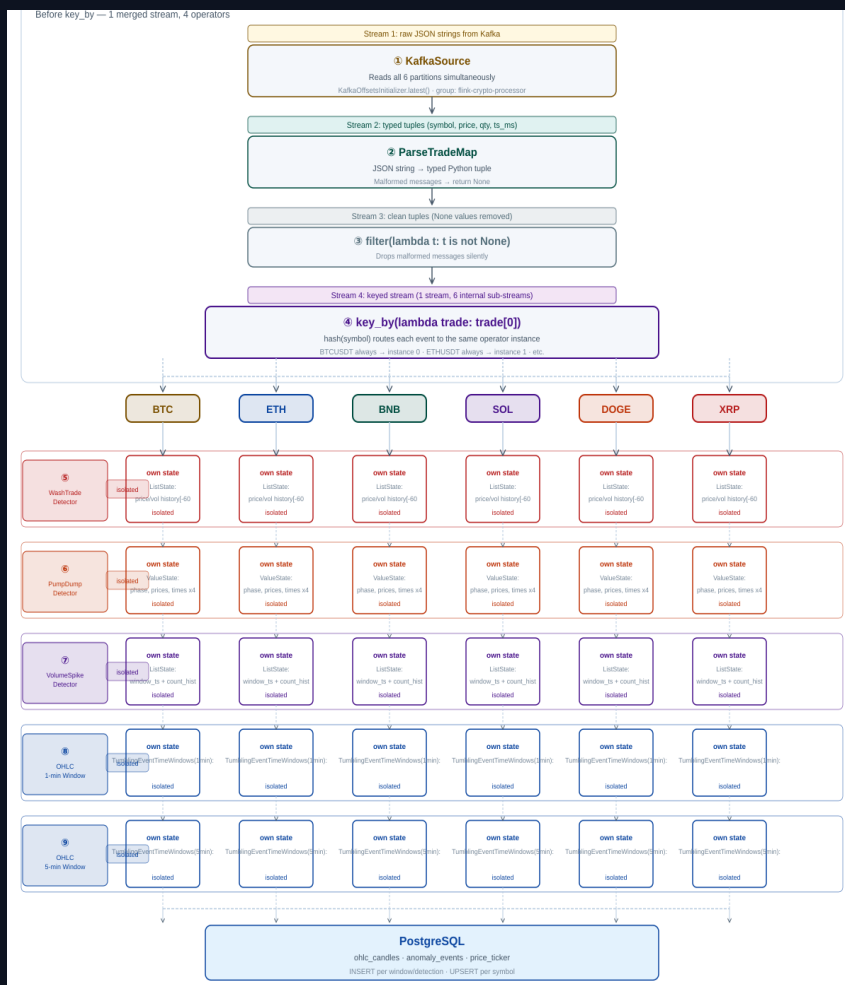


Kafka splits the stream into 6 lanes (one per coin). Trades for the same coin stay in order, the work runs in parallel, and we can scale up later with no code change.

INSIDE FLINK

How the pipeline splits into streams and operators

4 streams · 9 operators in code · 30 isolated operator instances at runtime (5 operators × 6 symbols)



After `key_by(symbol)`, each coin gets its own isolated state in every Flink operator. BTCUSDT's price history never mixes with DOGEUSDT's price history — guaranteed by Flink.

Why this matters for correctness:

Without isolation, BTC's \$64,000 price and DOGE's \$0.086 price would be in the same list. The swing calculation would show 74,000% — every trade would fire a `WASH_TRADE` alert. With `key_by`, each coin only sees its own history.

The three cheating patterns we detect

~ Wash trade

What it is:

A person quietly buys and sells to themselves to make a coin look busier than it really is.

Detection:

Price swings $\pm 0.3\%$ AND volume is 1.5× the rolling average across last 60 trades.

Flink state:

ListState — rolling price and volume history, capped at 60 entries per coin.

Pump & dump

What it is:

A group pushes a price up fast, then sells everything at the top, leaving buyers with losses.

Detection:

Price rises 0.4%, then drops 0.3% from the peak within 4 minutes.

Flink state:

ValueState — 3-phase state machine:
WATCHING → PUMP_CONFIRMED → ALERT.

Volume spike

What it is:

A sudden flood of trades — can signal the beginning of a flash crash or manipulation.

Detection:

Trade count in 10 seconds exceeds 2× the 10-minute rolling average. 30s cooldown.

Flink state:

ListState — sliding 10s window + 60-bucket rolling history per coin.

INSIDE POSTGRESQL

3 Tables on PostgreSQL

```
$ docker exec postgres psql -U flink -d crypto_market -c "\dt"
```

List of relations

Schema	Name	Type	Owner
public	anomaly_events	table	flink
public	ohlcv_candles	table	flink
public	price_ticker	table	flink

(3 rows)

```
$ docker exec postgres psql -U flink -d crypto_market -c \  
"SELECT symbol, pattern_type, severity, description FROM anomaly_events \  
WHERE detected_at >= NOW() - INTERVAL '1 hour' ORDER BY detected_at DESC LIMIT 10;"
```

symbol	pattern_type	severity	description
SOLUSD	VOLUME_SPIKE	MEDIUM	36 trades in last 10s vs avg 17.5 (2.1x normal - 10-min rolling baseline)
SOLUSD	VOLUME_SPIKE	MEDIUM	51 trades in last 10s vs avg 25.4 (2.0x normal - 10-min rolling baseline)
XRPUSD	VOLUME_SPIKE	MEDIUM	34 trades in last 10s vs avg 16.9 (2.0x normal - 10-min rolling baseline)
DOGEUSD	VOLUME_SPIKE	MEDIUM	37 trades in last 10s vs avg 18.2 (2.0x normal - 10-min rolling baseline)
DOGEUSD	VOLUME_SPIKE	MEDIUM	18 trades in last 10s vs avg 8.7 (2.1x normal - 10-min rolling baseline)
XRPUSD	VOLUME_SPIKE	MEDIUM	20 trades in last 10s vs avg 9.7 (2.1x normal - 10-min rolling baseline)
BNBUSDT	VOLUME_SPIKE	MEDIUM	50 trades in last 10s vs avg 24.9 (2.0x normal - 10-min rolling baseline)
BNBUSDT	VOLUME_SPIKE	MEDIUM	46 trades in last 10s vs avg 23.0 (2.0x normal - 10-min rolling baseline)
DOGEUSD	VOLUME_SPIKE	MEDIUM	41 trades in last 10s vs avg 20.1 (2.0x normal - 10-min rolling baseline)
SOLUSD	VOLUME_SPIKE	MEDIUM	52 trades in last 10s vs avg 25.9 (2.0x normal - 10-min rolling baseline)

(10 rows)

```
$ docker exec postgres psql -U flink -d crypto_market -c \  
"SELECT symbol, window_size, open_price, high_price, low_price, close_price, trade_count \  
FROM ohlc_candles ORDER BY created_at DESC LIMIT 10;"
```

symbol	window_size	open_price	high_price	low_price	close_price	trade_count
SOLUSD	1min	74.0999984741211	74.13999938964844	74.08000183105469	74.12999725341797	647
XRPUSD	5min	1.0479999780654907	1.0479999780654907	1.0467000007629395	1.0477999448776245	822
BTCUSD	1min	59489.98046875	59536.71875	59484.01171875	59529.828125	2622
SOLUSD	5min	74.18000030517578	74.18000030517578	74.05999755859375	74.12999725341797	3457
XRPUSD	1min	1.0472999811172485	1.0477999448776245	1.0471999645233154	1.0477999448776245	165
DOGEUSD	5min	0.07240000367164612	0.07242999970912933	0.07231000065803528	0.07242999970912933	798
DOGEUSD	1min	0.07237999886274338	0.07242999970912933	0.07237999886274338	0.07242999970912933	216
BNBUSDT	5min	552.6400146484375	552.6400146484375	552	552.3599853515625	1099
BNBUSDT	1min	552.1099853515625	552.3699951171875	552.0999755859375	552.3599853515625	205
ETHUSD	5min	1593.81005859375	1593.81005859375	1591.1500244140625	1593.3399658203125	5836

(10 rows)

```
$ docker exec postgres psql -U flink -d crypto_market -c \  
"SELECT symbol, price, price_change_pct, volume_1min, updated_at \  
FROM price_ticker ORDER BY symbol;"
```

symbol	price	price_change_pct	volume_1min	updated_at
BNBUSDT	552	-0.036220971135411936	40.55600031418726	2026-06-30 07:53:05.54359+00
BTCUSD	59496	-0.005042101547925175	14.60370994639743	2026-06-30 07:53:05.511159+00
DOGEUSD	0.0723700001835823	0.02763974781098655	164660	2026-06-30 07:53:05.53067+00
ETHUSD	1592.4300537109375	0.009421986792659098	238.11070095554896	2026-06-30 07:53:05.538231+00
SOLUSD	74.08000183105469	-0.05396777394100725	9608.975996589288	2026-06-30 07:53:05.518906+00
XRPUSD	1.0472999811172485	0.028646662658107433	33610.00028092414	2026-06-30 07:53:05.52474+00

(6 rows)

WHAT THE USER SEES

The live dashboard

Five live panels

Price ticker
price

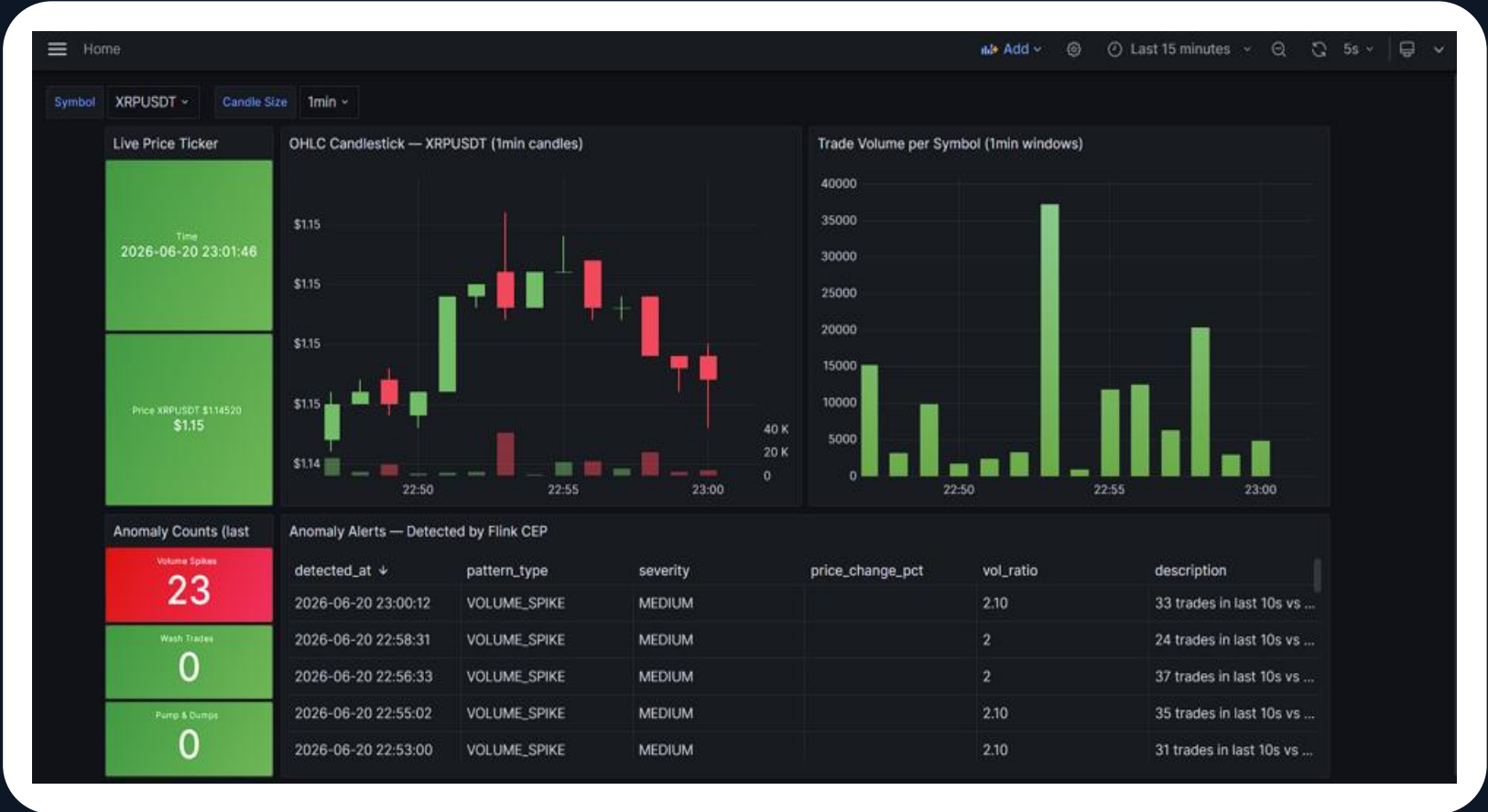
Candles
1-min or 5-min

Volume bars
per coin

Alert count
last hour

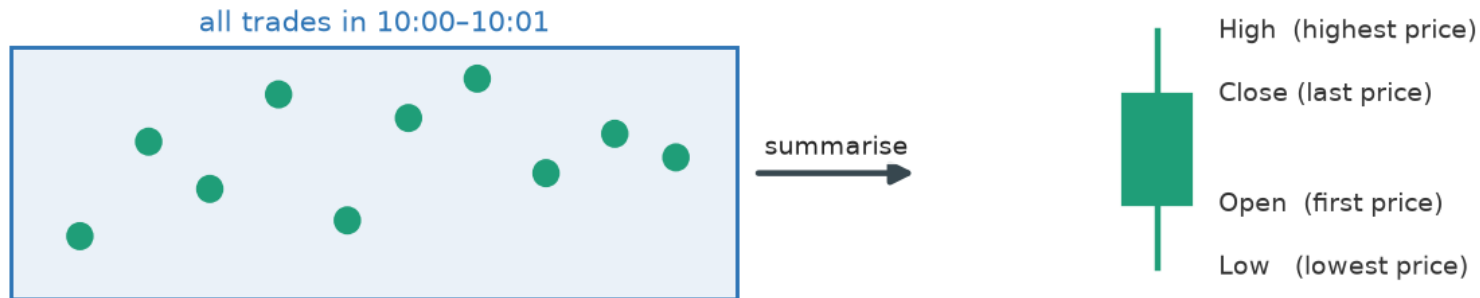
Alerts table
what + how serious

Pick a coin and candle size from the menus.



Turning trades into price candles

Flink turns 1 minute of trades into one candle



This 'OHLC candle' is one bar on a trading chart. We also make a 5-minute version.

Flink groups one minute of trades into a single candle: first, highest, lowest and last price. It uses each trade's real time, so late trades still land in the right candle.

IDEA 2

Turning trades into price candles

Traders read price candles. A candle is a simple summary of a short period: the first price (open), the highest, the lowest, and the last price (close). Flink groups all trades in each window into a single candle using `TumblingEventTimeWindows`.

1-min window

open · high · low · close · volume · count

→ *Grafana candlestick chart*

5-min window

open · high · low · close · volume · count

→ *Broader trend analysis*

The Grafana dashboard has a `$candle_size` dropdown — switch between 1-minute and 5-minute candles without any code change.

Green candle

Close price > open price — buyers were in control that minute

Red candle

Close price < open price — sellers were in control that minute

Getting it right

We use the trade's exchange timestamp, not arrival time. A 3-second watermark absorbs late messages. A 10-second idleness setting stops a quiet coin from freezing all windows.

Let's watch it run

1

`docker compose up -d --build`

Six parts start together. Kafka and PostgreSQL pass health checks first.

2

`http://localhost:3000` → `admin / admin`

Dashboard auto-loads from provisioning files. Select DOGEUSDT from the dropdown.

3

Switch `$symbol` dropdown → DOGEUSDT

DOGE fires anomaly alerts fastest. Watch the red alert count climb.

4

Switch `$candle_size` → 1min or 5min

All panels update instantly — candlestick, volume bars, alerts table.

5

`http://localhost:8080` → Kafka UI

Shows raw-trades topic, 6 partitions, live messages flowing in real time.

LESSONS LEARNED

What worked, what was hard, what's next

Worked well

- Simple, well-known design (Kafka→Flink→DB→dashboard)
- One command starts everything — easy to try
- One lane per coin keeps data tidy and easy to scale

Was hard

- Parts start together, but Kafka needs a moment — had to wait
- A quiet coin froze the others until one setting fixed it
- Tuning the rules to show alerts fast in a short demo

Next

- Use Flink SQL to write less code
- Run on a bigger cluster for more coins
- Let it learn each coin's 'normal' automatically

Thank you

Live crypto trades → cheating alerts on a dashboard, in seconds — started with one command.

Kodanda Challa · Real-Time Big Data Processing 2025–2026

Questions welcome.

Standard course shape: collect → Kafka → Flink → database → dashboard. Every box is one Docker container.

